Bolted Flange Joint Report

SPARK-2 MCC–Injector/Faceplate Flange Joint

Summary of joint stiffness split, preload, peak bolt force, and quick safety indicators.

Calculation Trace

Pc = 500 psi -> Pc = 3.44738 MPa

D_{eff} = 5 in -> D_{eff} = 127 mm

A = (π/4)·D_{eff}^2 = (π/4)·(127)^2 = 12667.7 mm^2

P_{total} = Pc·A = (3.44738)·(12667.7) = 43670.3 N

P_{per}_bolt = P_{total}/N_b = (43670.3)/8 = 5458.79 N

$k_b = (A_d \cdot A_t \cdot E_b)/(A_d \cdot l_t + A_t \cdot l_d)$ = (50.3·36.6·200000)/(50.3·10 + 36.6·20) = 298134 N/mm

$k_{frustum}$(Flange stack (symmetric)) x2: t=6, D=12 -> 1.36719e+06 N/mm each

$k_m = 1 / \Sigma(1/k_i)$ (series combination of frusta)

k_m = 683594 N/mm

$C = k_b/(k_b+k_m)$ = 298134/(298134+683594) = 0.303683

NASA-STD-5020 App A preload bounds (PDF p.41):

Ppi_{min} = Ppi·(1−Γ_p) = 12000·(1−0.2) = 9600 N

Ppi_{max} = Ppi·(1+Γ_p) = 12000·(1+0.2) = 14400 N

ΔF_{moment} = 0 (moment disabled)

F_b_max = Ppi_{max} + C·P_{per}_bolt + ΔF_{moment} = 14400 + 0.303683·5458.79 + 0 = 16057.7 N

F_{proof}_allow = $S_p \cdot A_t$ = 600·36.6 = 21960 N

MS_{proof} = (F_{proof}_allow/F_b_max) − 1 = (21960/16057.7) − 1 = 0.367565

NASA-STD-5020 separation requirement (PDF p.42, Eq A.2-3b):

L_{min}_per_{bolt} = L_{min}_total/N_b = 0/8 = 0 N

Ppi_{req}_min = (1−C)·P_{per}_bolt + ΔF_{moment} + L_{min}_per_{bolt} = (1−0.303683)·5458.79 + 0 + 0 = 3801.05 N

FoS_{sep} = Ppi_{min}/Ppi_{req}_min = 9600/3801.05 = 2.52562

MS_{sep} = FoS_{sep} − 1 = 1.52562

Inputs

Input	Value
Pc (psi)	500.00
Effective pressure diameter (in)	5.0000
Bolt count (N_b)	8
Bolt nominal diameter d (mm)	8.0000
Clearance hole diameter d_h (mm)	8.5000
A_t (mm^2)	36.60
A_d (mm^2)	50.30

Input	Value
Bolt modulus E_b (MPa)	200,000.0
Threaded length l_t (mm)	10.00
Unthreaded length l_d (mm)	20.00
Proof strength S_p (MPa)	600.00
Ppi_{nom} (N) (per bolt)	12,000.0
Γ_p (preload variation factor)	0.2000
L_{min}_total (N)	0.0000
Target separation factor	2.5000
Member layers	Flange stack (symmetric) (t=12.0mm, E=71000.0MPa, D=12.0mm)

Computed results

Quantity	Value
P_{total} (N)	43,670.3
P_{per}_bolt (N)	5,458.8
k_b (N/mm)	298,134.4
k_m (N/mm)	683,593.9
$C = k_b/(k_b+k_m)$	0.3037
Ppi_{nom} (N)	12,000.0
Ppi_{min} (N)	9,600.0
Ppi_{max} (N)	14,400.0
ΔF_{moment} (N)	0.0000
F_b_max (N)	16,057.7
F_{proof}_allow (N) = $S_p \cdot A_t$	21,960.0
MS_{proof} = (Allow/Applied) – 1	0.3676
Ppi_{req}_min (N)	3,801.0
$FoS_{sep} = Ppi_{min}/Ppi_{req}$_min	2.5256
$MS_{sep} = FoS_{sep} - 1$	1.5256
Required bolts for target separation (estimate)	8

Checks

Check	Status
Proof (MS_proof >= 0)	PASS

Check	Status
Separation (FoS_sep >= target=2.50)	PASS
Sanity: C <= 0.6	PASS

Notes

Preload bounds and separation requirement follow NASA-STD-5020 Appendix A (A.2.1): Ppi_min/Ppi_max per Eq. (A.2-1)/(A.2-2) and Ppi_req_min per Eq. (A.2-3b). Joint stiffness split (C) uses a Shigley-style stiffness model.

Generated by the bolted flange calculator.